SIMATS ENGINEERING
ASSESSMENT TOOL 1

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REG NO:192521375
COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0703

COURSE OUTCOME COVERED

CO5:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2, BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3 **Total Marks:30**

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability

2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

1. Remote Rural Branch Office Network Design

Problem Understanding & Constraints

The company requires:

- High-speed internet in a rural area without wired broadband.
- Cloud application support.
- Video conferencing.
- Secure communication with headquarters.
- Limited budget.
- High reliability with minimal downtime.
- Future scalability.

Problem Decomposition

1. Select internet connection.
2. Choose networking devices.
3. Implement security mechanisms.
4. Ensure scalability.

Solution Design

Component	Recommendation	Justification
Internet Connection	5G Fixed Wireless Access (Primary) + Satellite Internet (Backup)	5G offers high speed at lower cost. Satellite ensures connectivity during outages.
Router	Enterprise Router with Dual-WAN	Supports automatic failover between 5G and satellite.
Switch	Gigabit Managed Switch	Efficient LAN connectivity and VLAN support.
Wireless	Wi-Fi 6 Access Points	Better speed, capacity, and future readiness.
Firewall	Next-Generation Firewall (NGFW)	Protects against cyber threats.
VPN	IPSec Site-to-Site VPN	Secure communication with headquarters.
Power Backup	UPS	Prevents downtime during power failures.

Application of Constraints

- Bandwidth: Wi-Fi 6 optimizes wireless performance.
- Reliability: Dual internet links provide redundancy.
- Cost: 5G is cheaper than laying fiber.
- Scalability: Additional APs and bandwidth can be added later.

Logical Reasoning

Using 5G as the primary connection reduces installation costs while providing sufficient speed for cloud applications. Satellite backup guarantees uninterrupted business operations during network failures.

Final Recommendation

Deploy 5G Fixed Wireless Access + Satellite Backup, Enterprise Router, Managed Switch, Wi-Fi 6 APs, NGFW, IPSec VPN, and UPS for a secure, reliable, and scalable network.

2. University Campus Wireless Network Design

Problem Understanding & Constraints

Requirements:

- More than 3,000 simultaneous users.
- Coverage across classrooms, labs, library, and hostels.
- Limited number of access points.
- Secure access.
- Minimal interference.
- Budget constraints.

Problem Decomposition

1. Access point placement.
2. Frequency band selection.
3. Authentication.
4. Performance optimization.

Solution Design

Component	Recommendation	Justification
Access Points	Wi-Fi 6 Access Points	Supports many concurrent users.
Placement	Ceiling-mounted APs in central locations	Maximizes coverage with fewer APs.
Frequency	5 GHz for classrooms/labs, 2.4 GHz for hostels	5 GHz reduces interference; 2.4 GHz provides wider coverage.
Roaming	Fast Roaming (802.11r)	Smooth movement between APs.
Authentication	WPA3-Enterprise with RADIUS Server	Secure login using university credentials.
VLANs	Separate Student, Faculty, and Guest Networks	Improves security and traffic management.
Load Balancing	Enable AP Load Balancing	Distributes users evenly across APs.

Access Point Placement

- One AP for every 2–3 classrooms.
- One AP per laboratory.
- Multiple APs in the library.
- Corridor-mounted APs in hostels.
- Outdoor APs for open campus areas.

Application of Constraints

- **Coverage:** Strategic AP placement minimizes hardware requirements.
- **Performance:** Wi-Fi 6 efficiently handles high user density.
- **Security:** WPA3-Enterprise protects user authentication.
- **Cost:** Optimized AP placement reduces deployment expenses.

Logical Reasoning
Using Wi-Fi 6 allows more simultaneous connections without deploying excessive APs. Dual-band operation minimizes interference while maximizing coverage.

Final Recommendation
Deploy strategically placed **Wi-Fi 6 APs**, use **5 GHz** in high-density areas, **2.4 GHz** where extended coverage is needed, implement **WPA3-Enterprise with RADIUS**, and separate users using VLANs.

3. Secure Financial Institution Network Design

Problem Understanding & Constraints
Requirements:

- Protect sensitive customer information.
- Secure online banking services.
- Comply with security regulations.
- Maintain high performance.
- Minimize operational costs.

Problem Decomposition

1. Network segmentation.
2. Firewall deployment.
3. Authentication.
4. Intrusion detection.
5. Compliance.

Solution Design

Component	Recommendation	Justification
Network Segmentation	VLANs + DMZ	Separates banking servers, employee systems, and public services.
Firewall	Next-Generation Firewall	Filters malicious traffic and supports application awareness.

Authentication	Multi-Factor Authentication (MFA) + Active Directory	Strong identity verification.
IDS/IPS	Intrusion Detection and Prevention System	Detects and blocks cyber-attacks.
VPN	SSL/IPSec VPN	Secure remote employee access.
Endpoint Protection	Endpoint Detection and Response (EDR)	Protects employee devices.
Monitoring	SIEM System	Centralized security monitoring and compliance reporting.

Application of Constraints

- **Security:** VLANs isolate critical systems.
- **Performance:** Internal traffic remains fast through switching.
- **Compliance:** MFA, logging, and SIEM support regulatory requirements.
- **Budget:** Virtual VLANs reduce the need for additional physical hardware.

Logical Reasoning

Separating banking servers from employee networks reduces attack risks. NGFW and IDS/IPS provide layered protection while maintaining network performance. MFA prevents unauthorized access without significantly increasing operational costs.

Final Recommendation

Implement VLAN-based segmentation, DMZ, NGFW, MFA, IDS/IPS, VPN, EDR, and SIEM to achieve a secure, compliant, high-performance, and cost-effective financial network.